



COMPUTER NETWORK AND INFORMATION SECURITY — FINAL PRESENTATION

Post-Quantum Cryptography

Securing tomorrow's data against today's harvested traffic

Instructed by: Assoc. Prof. Suratosse Tritilanunt, Ph.D

Submitted by: Nathan Tanaboriboon (ID: 6681224)

The Trust Behind Every Message

Networks move data; cryptography keeps it confidential, unaltered, and exclusive to the intended receiver.



Confidentiality

Data is encrypted so interceptors along the path cannot read it.



Integrity

Tampering or corruption in transit can be detected.



Mathematical Trust

Security rests on problems that are easy to set, hard to reverse — without the key.

Today's puzzle: every standard in use — AES, RSA, ECC — is built on math problems that classical computers cannot solve in practical time (decades to millennia). Quantum computers threaten to change that timeline dramatically.

Symmetric Cryptography: One Shared Key



Analogy: Lock a message in a chest, send the chest, then send the key through a separate, more secure channel. One key both locks and unlocks.

AES — Advanced Encryption Standard

- Operates on 128-bit blocks across multiple rounds: SubBytes, ShiftRows, MixColumns, AddRoundKey
- Creates an avalanche effect — one bit change randomizes the entire output
- Deployed as AES-128 or AES-256
- No known shortcut beats brute force against AES-256 classically

Why It's Strong (Today)

Brute-forcing AES-256 requires astronomically more attempts than there are atoms to spend on the search — secure against any classical supercomputer in existence.

Shared-key model: fast, efficient — but key distribution is the weak link.

Asymmetric Cryptography: Public & Private Keys

A public key locks; only the matching private key can unlock. Security relies on “trapdoor functions” — easy forward, infeasible in reverse.



RSA

Integer Factorization Problem

Multiplying two huge primes is easy; factoring the result back apart is computationally infeasible for classical machines.

Analogy

Public key = a recipe for mixing two paint colors together. Anyone can mix paint that way. Private key = the only formula to separate the mixed color back into its originals, drop by drop.



ECC

Discrete Logarithm Problem

Tracks points across elliptic curves ($y^2 = x^3 + ax + b$) through repeated point addition — same security as RSA with far smaller keys.

Analogy

A billiard ball is struck N times on a warped, curved table and lands somewhere unpredictable. Public key = where it stops. Private key = how many times it was struck.

The Quantum Threat: A New Kind of Computation

Classical bits are 0 or 1. Quantum bits (qubits) exploit physics itself to search vastly more possibilities at once.



Superposition

A qubit holds a linear combination of 0 and 1 at once, letting n qubits represent 2^n states simultaneously.



Entanglement

Linked qubits update each other's state instantly, regardless of distance, coordinating computation.



Quantum Interference

Probability waves are manipulated to cancel wrong answers and amplify the correct one.

Analogy: a classical computer solves a maze by trying one path at a time, backtracking on dead ends. A quantum computer floods the entire maze with water at once, instantly revealing which path flows out.



Shor's Algorithm: Breaking RSA & ECC

Peter Shor, 1994 — uses the Quantum Fourier Transform to solve integer factorization and discrete logarithms in minutes, not millennia.

Impact: completely undermines asymmetric cryptography (RSA, ECC) — an attacker could derive private keys directly from public keys.

Analogy: Finding a Pattern in a Massive Sound Wave

Imagine a sound wave with a repeating pattern that recurs only once every few billion beats. A classical computer must count beats one at a time — unrealistically slow. A quantum computer instead floods all beats into qubits simultaneously (superposition); through the Quantum Fourier Transform, the irrelevant noise cancels itself out while the true repeating pattern amplifies. Only the answer survives. RSA and ECC's underlying math problems are exactly this kind of hidden pattern.



Grover's Algorithm: Weakening AES

Lov Grover, 1996 — a quadratic speedup for unstructured search, applied to brute-forcing symmetric keys.

Analogy: The Wall of Lockers

With N lockers and one hidden object, checking each one by one is too slow. Grover's algorithm peeks into all lockers at once; with each pass, Amplitude Amplification fades out the empty lockers' probability and boosts the correct one — until only it remains.

Result: an N -locker brute force search collapses to roughly $\sqrt{N}$ attempts.

AES-128

64

effective bits of security — vulnerable

AES-256

128

bits remain — still a strong quantum-resistant barrier



“Harvest Now, Decrypt Later”

The immediate crisis: today's encrypted traffic is already being stockpiled.

Analogy

Even an uncrackable chest can be copied. The corrupted mail company can't open it today — but it photographs and warehouses every chest, betting that a future tool (a quantum computer) will eventually unlock the whole archive at once.



Intercept & Archive

Threat actors and nation-states are already capturing and storing large volumes of encrypted traffic.



Protected... For Now

Classical RSA/ECC encryption holds today, but the ciphertext sits in repositories, waiting.



Retroactive Decryption

Once a stable quantum computer runs Shor's algorithm, decades of stored communication can be decrypted at once.

The Solution: Post-Quantum Cryptography

PQC vs QKD: Quantum Key Distribution (QKD) needs dedicated quantum hardware to work. Post-Quantum Cryptography (PQC) is purely software — classical math algorithms that run on today's ordinary computers, no hardware changes required.



Lattice-Based Cryptography

Replaces prime factorization with the Shortest Vector Problem across noisy, multi-dimensional grids (Learning With Errors).

Analogy: a 500-dimensional forest of gridded trees. Your private key is a compass that filters static instantly. The public key is a book of noisy coordinates — anchored to the compass but looking like chaos to anyone else. A quantum attacker without the compass must test trillions of paths and gets lost.



Hash-Based Cryptography

Used for digital signatures. Links one-time signatures into Merkle Trees — security rests purely on one-way hash functions, no algebra to attack.

Analogy: a one-way meat grinder built into a tournament bracket. Secret phrases (private key) get ground and paired upward until one master block remains (public key). Signing reveals one path up the bracket — verifiable, but the grinder can never run backward, even for a quantum computer.

Global Standardization: The NIST FIPS Standards

Three finalized Federal Information Processing Standards coordinate the worldwide migration.

Standard	Algorithm	Math Domain	Replaces / Target
FIPS 203	ML-KEM (Kyber)	Module-LWE Lattice	Key encapsulation — replaces RSA, ECDH
FIPS 204	ML-DSA (Dilithium)	Module-LWE Lattice	Digital signatures — replaces RSA-Sig, ECDSA
FIPS 205	SLH-DSA (SPHINCS+)	Stateless Merkle Hash	Backup signature scheme

Key Size Challenge: An ECC key is ~32 bytes; an ML-KEM key exceeds 1,000 bytes. Larger keys can fragment packets during handshakes and strain constrained devices like IoT sensors or smartcards.

Migration Strategy: Getting There Safely

Two core principles guide enterprise adoption of PQC.



Hybrid Deployments

Wrap payloads in both a classical algorithm (e.g., ECDH) and a post-quantum standard (e.g., ML-KEM) simultaneously.

An attacker must break both layers to ever access the data — a safety net during the transition.



Crypto-Agility

Re-engineer software to remove hard-coded dependencies on any single algorithm.

Modular abstractions let cryptographic protocols and parameters update via configuration — no system rewrite needed.



Conclusion

- Quantum computers offer immense processing potential — and threaten the cryptographic trust underpinning the modern internet.
- Shor's algorithm undermines RSA and ECC; Grover's algorithm halves AES's effective strength.
- “Harvest Now, Decrypt Later” makes this an active threat today, not a future one.
- Hybrid deployments and crypto-agility are the essential, immediate steps toward a quantum-safe infrastructure.

The transition to post-quantum cryptography is not optional — it is a matter of when, not if.